Anonymized Qualitative Content Analysis (Mayring)

Data Basis and Anonymization

- Source file: Survey of doctors on telemedicine experience.
- Material type: 15 structured interview/questionnaire transcripts (Q1-Q30).
- Anonymization actions performed:
  - Removed direct identifiers (names, exact clinic names, ages in profile headers).
  - Replaced geographic references with `[LOCATION]`.
  - Replaced doctor labels with participant IDs `P01` to `P15`.
  - Removed duplicated transcript segment.
- Output corpus: `docs/survey_doctors_anonymized.txt`.

Methodological Approach (Mayring)

This analysis follows Mayring's qualitative content analysis using a **deductive-inductive structuring approach**:

1. **Definition of material**: Full anonymized corpus (15 participants, all 30 questions).
2. **Direction of analysis**: Identify perceived enablers, barriers, and implementation needs for telemedicine in clinical practice.
3. **Units of analysis**:
   - Coding unit: Meaningful statement/sentence in answer text.
   - Context unit: Full answer block for each question.
   - Evaluation unit: Participant-level transcript (`P01` to `P15`).
4. **Category development**:
   - Deductive categories from interview domains (usage, videocalls, scheduling, data handling, legal aspects).
   - Inductive refinement from recurring themes in statements.
5. **Revision and reduction**: Categories merged where semantically overlapping; frequencies counted at participant level.
6. **Interpretation**: Cross-category synthesis into implementation implications.

Category System (Final)

Main Category	Description	Participants (n=15)
C1 Infrastructure and access constraints	Internet instability, low digital literacy, lack of devices, rural access barriers	15

Main Category	Description	Participants (n=15)
C2 Informal platform dependence	Use of Telegram/WhatsApp/phone as de-facto telemedicine channels	15
C3 Telemedicine as supplementary care	Remote care useful for follow-up/triage but not equivalent to physical exam	15
C4 Documentation and workflow burden	Double documentation, paper fallback, manual archiving	15
C5 Data protection and privacy risk	Security concerns, leakage risk, personal-device vulnerability	15
C6 Legal and regulatory uncertainty	Unclear liability, unclear status of online advice/prescriptions	15
C7 Boundary strain and workload shift	After-hours messages, expectation of immediate response	10
C8 Demand for institutional solution	Need for national platform, standards, training, ministry guidance	15
C9 Uneven maturity by setting	Advanced digital use in a minority vs mostly low-maturity settings	15

Coding Rules and Anchor Examples

C1 Infrastructure and access constraints

- Rule: Code statements about connectivity, devices, digital literacy, or regional infrastructure limitations.
- Example: "The biggest issue is connectivity. Many patients have weak mobile signals or do not have smartphones."

C2 Informal platform dependence

- Rule: Code mentions of consumer apps as primary clinical communication tools.
- Example: "Not special medical apps, just Telegram."

C3 Telemedicine as supplementary care

- Rule: Code statements that define remote care as follow-up support rather than diagnostic replacement.
- Example: "It is good for monitoring and guidance but cannot replace a physical examination."

C4 Documentation and workflow burden

- Rule: Code references to manual notes, printing chats, dual systems, extra admin work.
- Example: "I spend extra time documenting things twice – once digitally and once on paper."

C5 Data protection and privacy risk

- Rule: Code explicit security, confidentiality, access control, data loss, unauthorized forwarding concerns.
- Example: "Telegram is convenient but not designed for medical data."

C6 Legal and regulatory uncertainty

- Rule: Code uncertainty on legality, liability, billing, prescribing, and formal recognition of teleconsultations.
- Example: "I'm always unsure whether it's legally safe to provide advice through Telegram."

C7 Boundary strain and workload shift

- Rule: Code blurred work/personal boundaries and after-hours demand.
- Example: "Patients often message late at night, expecting immediate replies."

C8 Demand for institutional solution

- Rule: Code explicit requests for official platform, standards, training, and ministry-led implementation.
- Example: "We need clear national regulations, institutional training, and official software approved by the Ministry of Health."

C9 Uneven maturity by setting

- Rule: Code variability in digital adoption between participants/settings.
- Example (high maturity): “40% of my patient interactions are conducted online.”
- Example (low maturity): “Never. We don’t have such a system.”

Case Summary (Cross-Category Interpretation)

1. **Systemic gap between demand and infrastructure**
   - Doctors see practical value in telemedicine (follow-up, triage, continuity), but implementation is constrained by connectivity and absent institutional platforms.
2. **Shadow digitalization is already happening**
   - Even where formal systems are missing, care has moved into consumer apps. This creates clinical utility but shifts risk to doctors and patients.
3. **Safety-governance mismatch**
   - Security and legal concerns are not peripheral; they are central adoption barriers. Doctors self-limit remote advice to avoid liability.
4. **Administrative paradox**
   - Digital communication can reduce patient travel but often increases clinician workload because documentation remains parallel (chat + paper).
5. **Two-speed adoption pattern**
   - A minority of participants show integrated digital workflows (scheduled video visits, EHR-linked communication), while most remain at informal, low-integration stages.

Mayring-Conformant Conclusion

From a Mayring perspective, the central content structure is clear: **telemedicine is perceived as useful and necessary, but currently practiced in a legally and technically under-institutionalized form**. The dominant latent pattern is not resistance to digital care itself, but resistance to **unsafe and unsupported digital practice conditions**.

Practical Implications for Your Thesis

- Prioritize a staged model: informal messaging -> secure communication -> integrated telemedicine platform.
- Treat legal framework and documentation automation as core implementation pillars, not secondary add-ons.
- Include rural connectivity and patient digital literacy as primary context variables in your discussion.
- Separate “clinical suitability” (follow-up vs diagnostics) by specialty in your interpretation chapter.

Files Produced

- Anonymized corpus: `docs/survey_doctors_anonymized.txt`
- Category frequency sheet: `docs/survey_mayring_category_counts.txt`
- This report: `docs/survey_doctors_mayring_analysis.md`